

Introduction to Fourier series and Transformation.

- FS expansion is used for periodic signals to expand them in terms of their harmonics which are sinusoidal and orthogonal to one another.

1. CT-CTFS

- periodic signal. 2. DT-DTFS

- for non periodic signal i.e. Real life signal
eg. Voice for analysis of non periodic
we need some tool which was given by
Joseph Fourier i.e. Fourier Transformation.
1. Continuous Time FT
2. Discrete Time FT.

FS and FT are used for analysis purposes.
- Laplace Transform design purposes (CT)
Z-Transform (DT)

They are used to model any arbitrary periodic signal with combination of sine & cosine.

Periodic signal is the signal which there is repetition of a particular structure from $-\infty$ to $+\infty$.

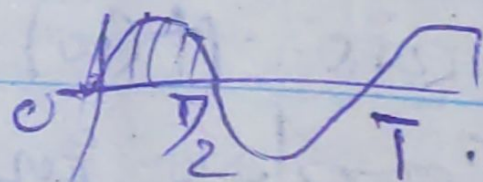
- is a signal that repeats itself at a particular time $\text{ft } x(t) \rightarrow \text{periodic signal}$.

$$x(t \pm T) = x(t)$$

Time period = $n(T_0)$ fundamental time period.
Small shift will not affect the whole signal.

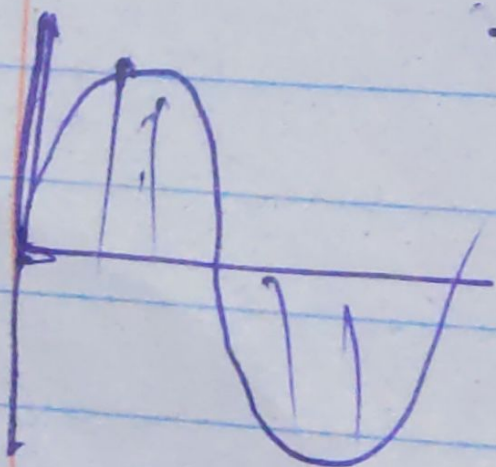
Harmonics - frequency pulse.

$f = \frac{\text{no of cycles}}{\text{Sec}}$



$T =$ fundamental time period of the signal function

F.S.



- Sinusoidal waveform.

- Cosine/sine wave.

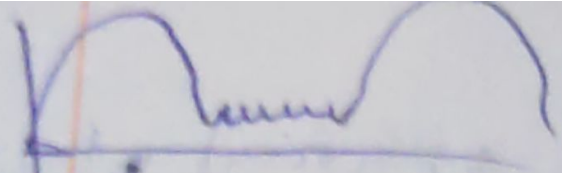
- not carrying information.

- Practically not a signal.

- We know everything about it.

- deterministic.

difference b/w FOS & FT.



FFT

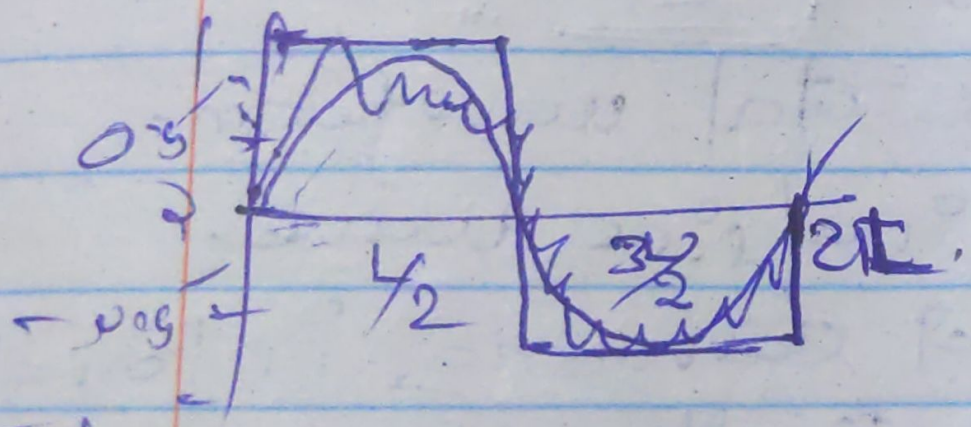
- different frequency.
- non-deterministic.
- Random.
- Carries information
- Carries something we don't know.

$$x(t) = 2 \sin t + \sin 2t - 7 \sin t$$

We need FT. in order to obtain the frequency response $H(\omega)$.

$H(\omega)$ is complex in nature.

$$|H(\omega)|^2 < H(\omega)$$



Different Fourier series expansion.

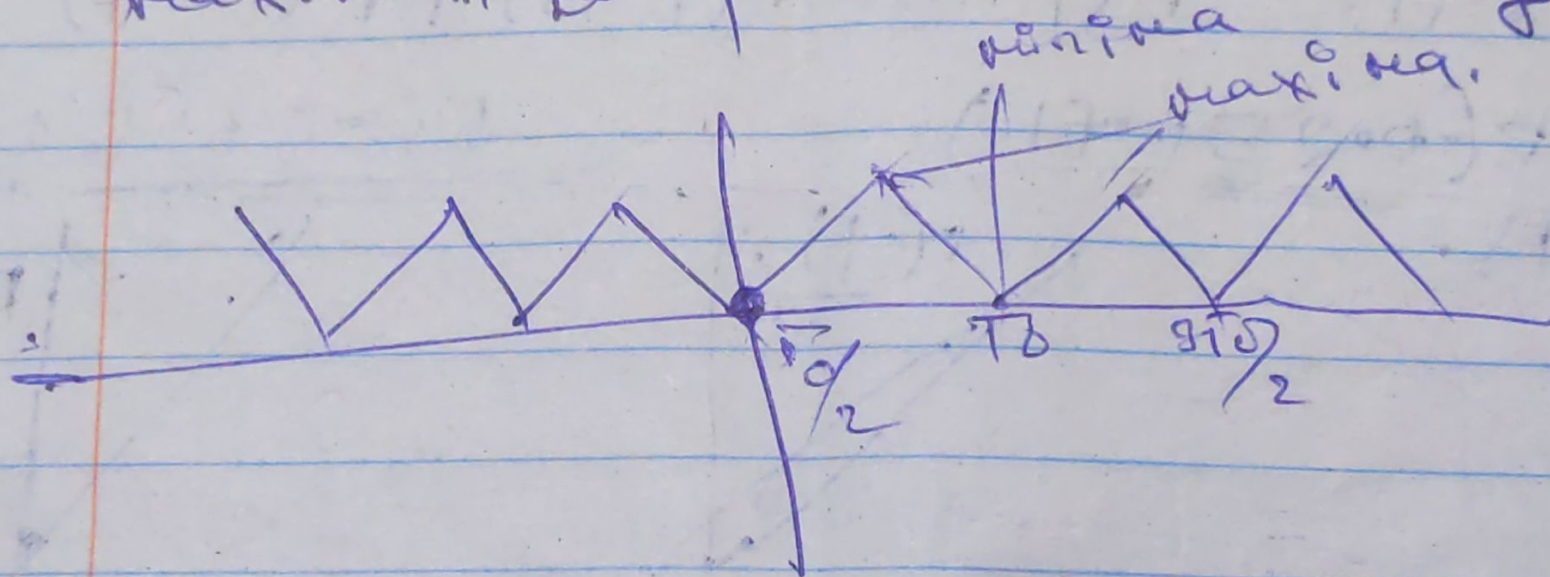
1. Trigonometric FT expansion
2. Complex
3. Polar (harmonic)

Condition -

1. not all periodic signal are FS and not all non-periodic signal are FS.

Ans. why.

Condition for periodic signal should have finite number of maximum and min over range period.



2. Periodic signal should have finite number of discontinuity over range of time period.

